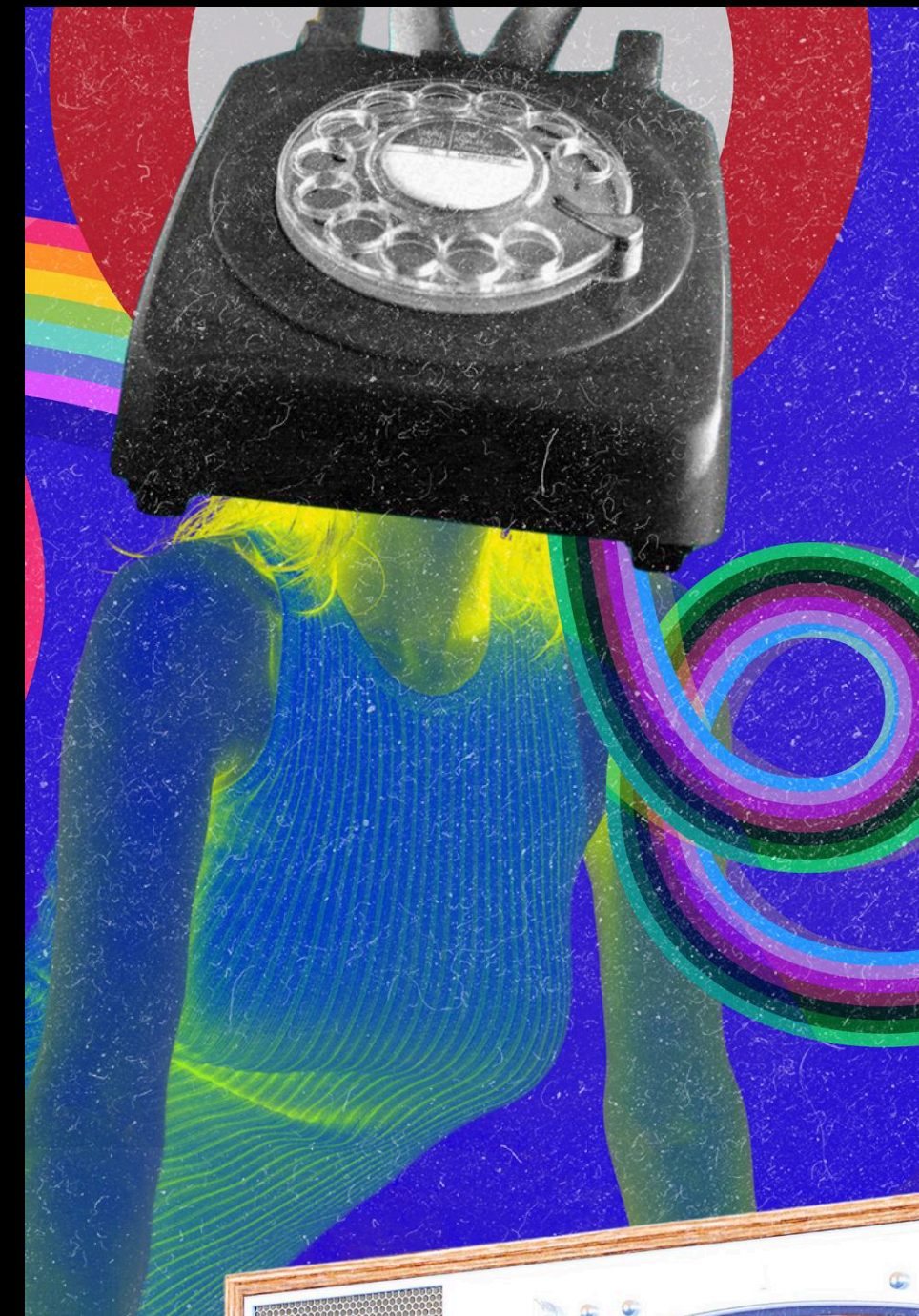


Assignment 5: Beyond the Screen

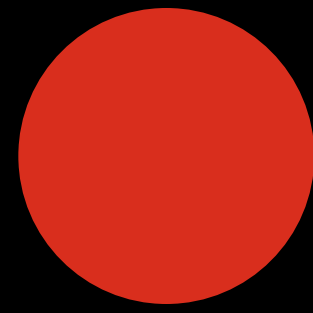


Defining the Device

Interaction Specifications

Wizard-of-Oz Prototype (v1)

Testing Results



Defining the Device

AirDJ

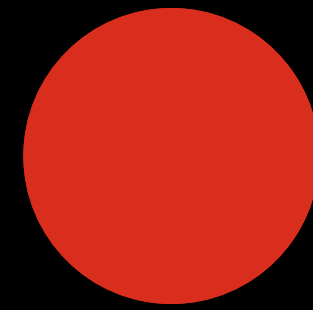
AirDJ is a screenless DJ performance system that frees performers from their booth.

It consists of four physical components: a wrist band that detects hand gesture, a mixer unit where DJ can turn on the AirDJ, a speaker unit that interprets the hand gestures and plays music, and a light system that responds to the music in real time. The system allows the DJ to step away from their station entirely, using hand gestures to control music playback while the venue lighting responds to the audio in real time.



CAPTION

CAPTION



Concept 1 + Concept 2

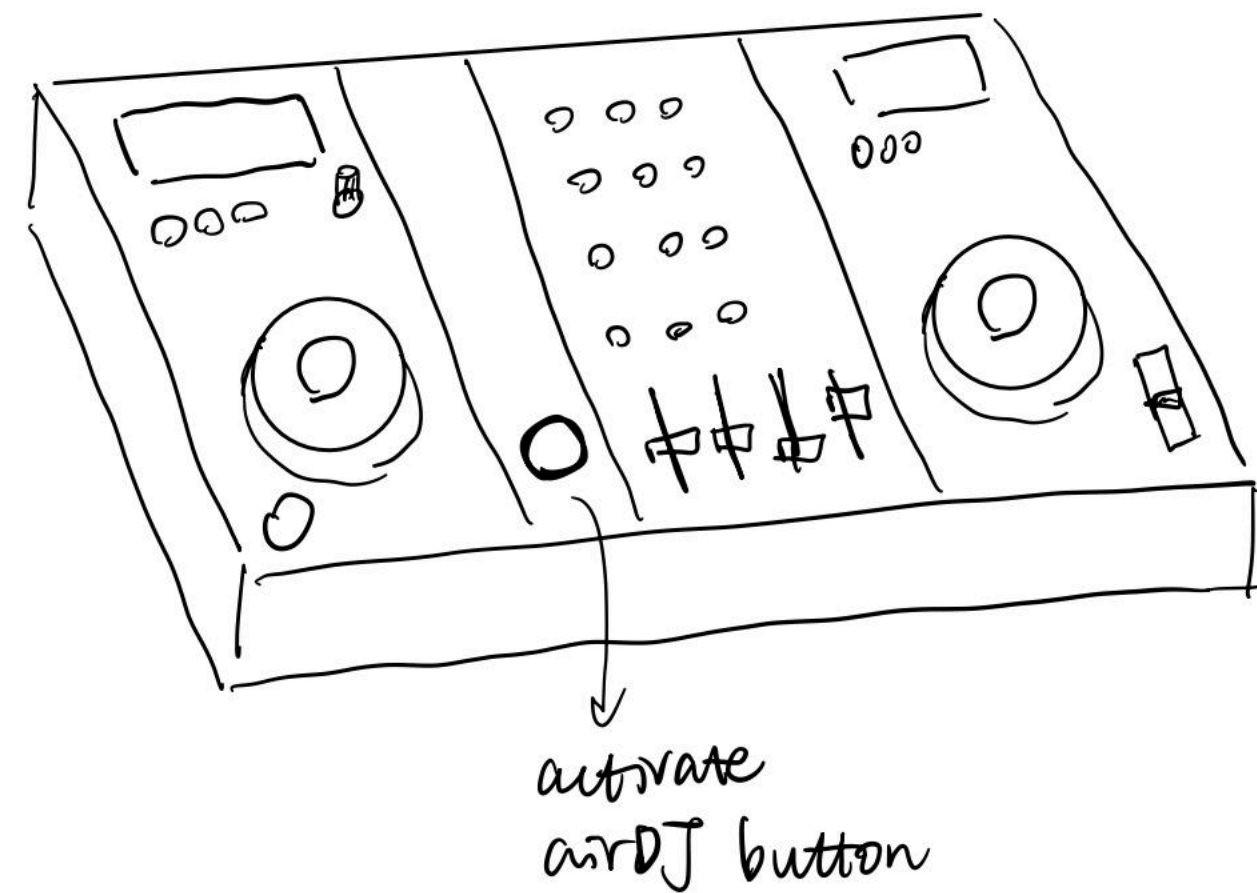
(Gestural input > audio output)

(audio output > light output)

Tempo Control Anywhere on the Stage

● CONCEPT 1

DJ MIXER



WEARABLE



Tempo Control Anywhere on the Stage

● CONCEPT 1

AIRDJ: AUDIO SYSTEM STORYBOARD - CONCEPT 1



The Status Quo (Constraint)

1. Traditional booth setup: DJ is stuck behind the workstation, limited in physical movement.



Desire for Movement (Problem)

2. DJ wants to move to hype the distant crowd, but cannot leave the mixer.



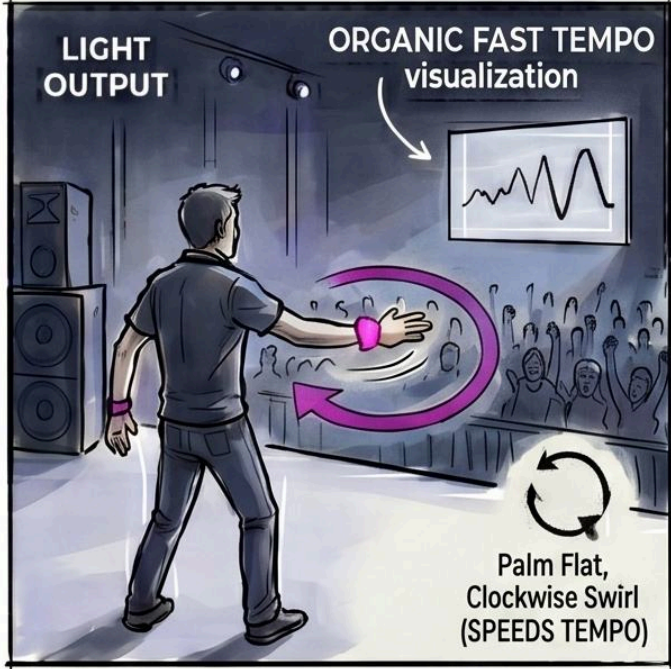
Connection/Activation (Action)

3. Activation: DJ hits the 'AIRDJ' button, establishing connection to the active wearable.



4. Stepping Away (Freedom)

4. Physical freedom: DJ walks away from the booth, moving to the front of the stage.



5. Screenless Control (Gesture Control)

5. Screenless Control: The DJ walks away from the station to the front of the stage. Standing free, they perform a flat-palm clockwise motion, causing the music to organically speed up.



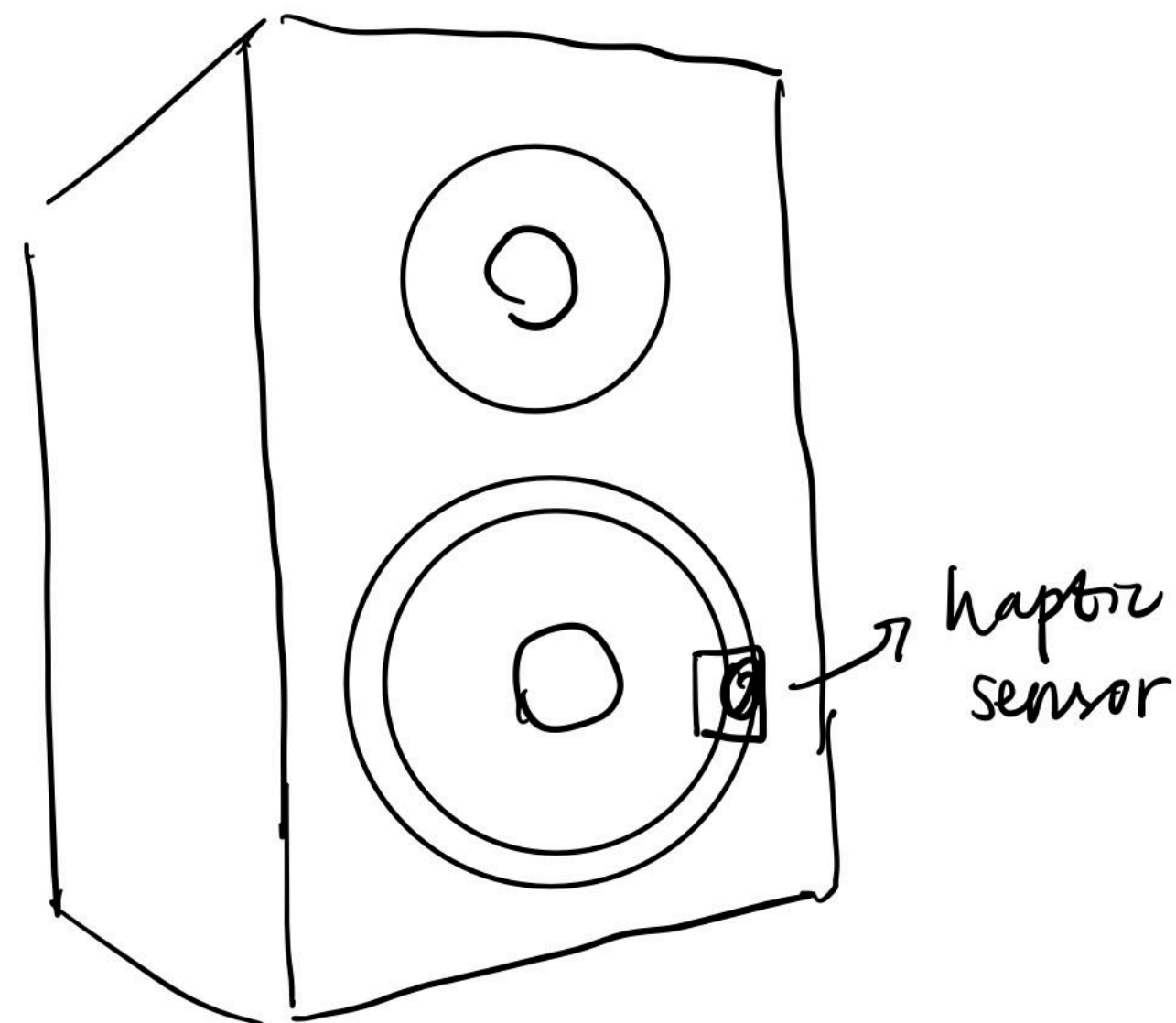
6. Dynamic Energy: (Audio response)

6. Dynamic energy: The music accelerates, its tempo tightening dramatically. The entire track becomes a high-tempo, intense, powerful auditory climax, creating a deep-tech rhythmic soundscape with rapid beats.

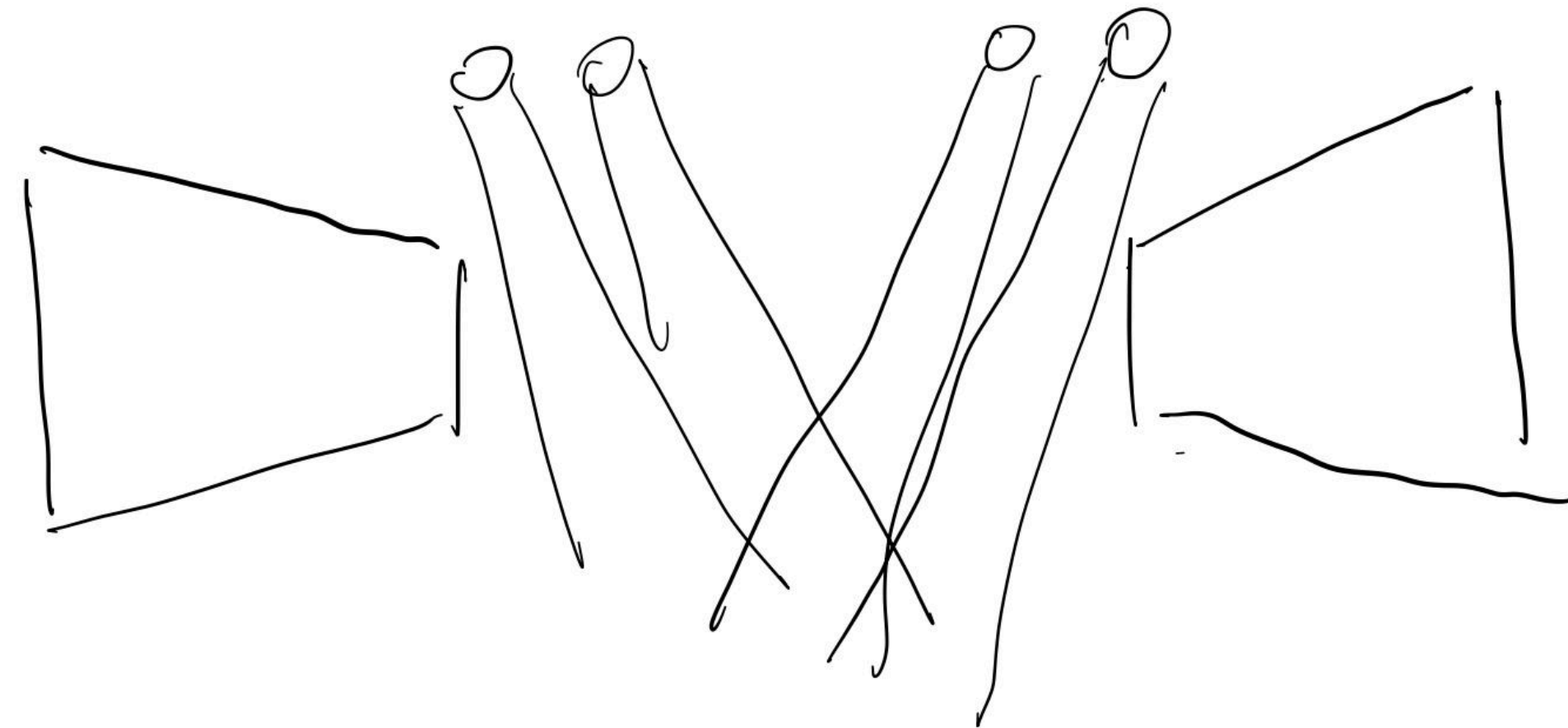
Sound-Reactive Ambient Lighting

● CONCEPT 2

SPEAKER



LIGHT SYSTEM



light system that changes color based on haptic and audio input (from the speaker)

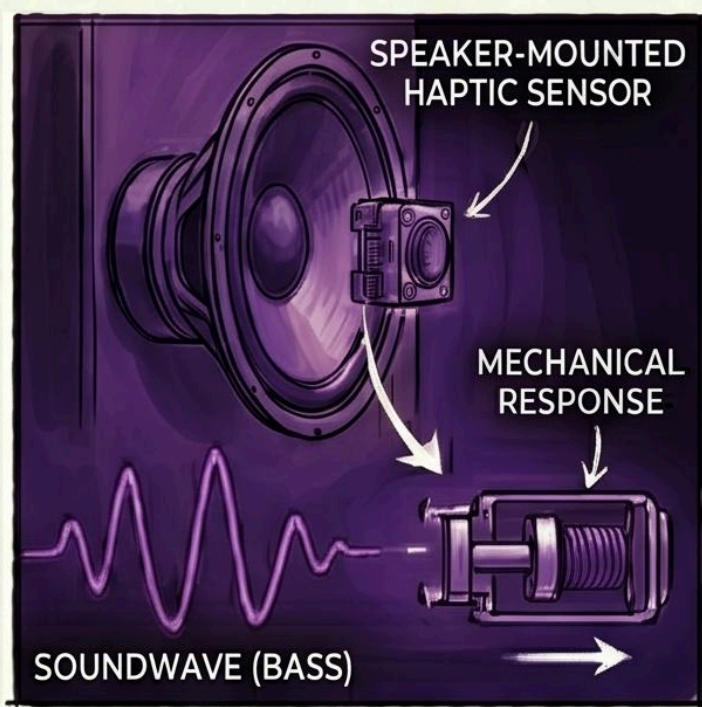
Sound-Reactive Ambient Lighting

HAPTIC BEAT-LIGHT MAPPING SYSTEM

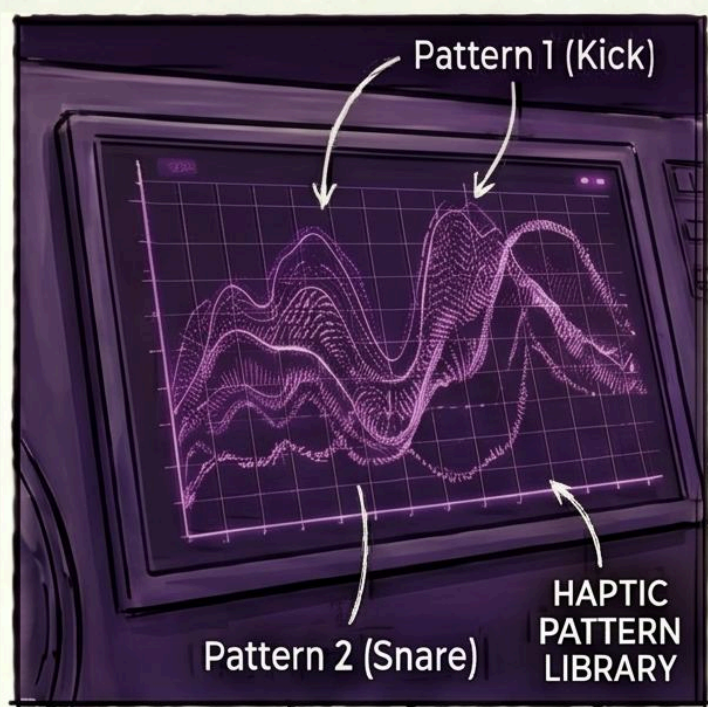
CONCEPT 2



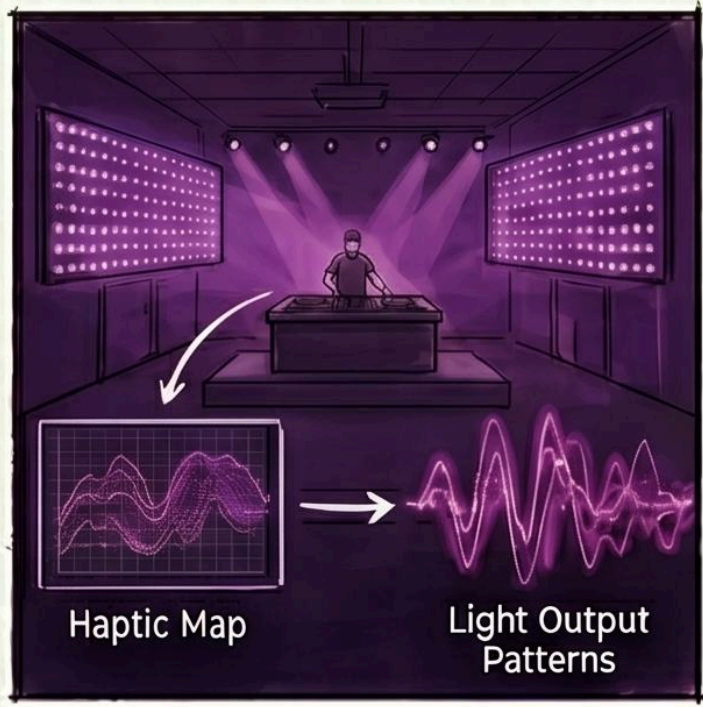
1. Initial Sensing (Calibration)
- The DJ plays a track. A light-sensitive haptic sensor, mounted on a speaker, analyzes the precise micro-vibrations and forces of the soundwaves, distinguishing between subtle haptic differences.



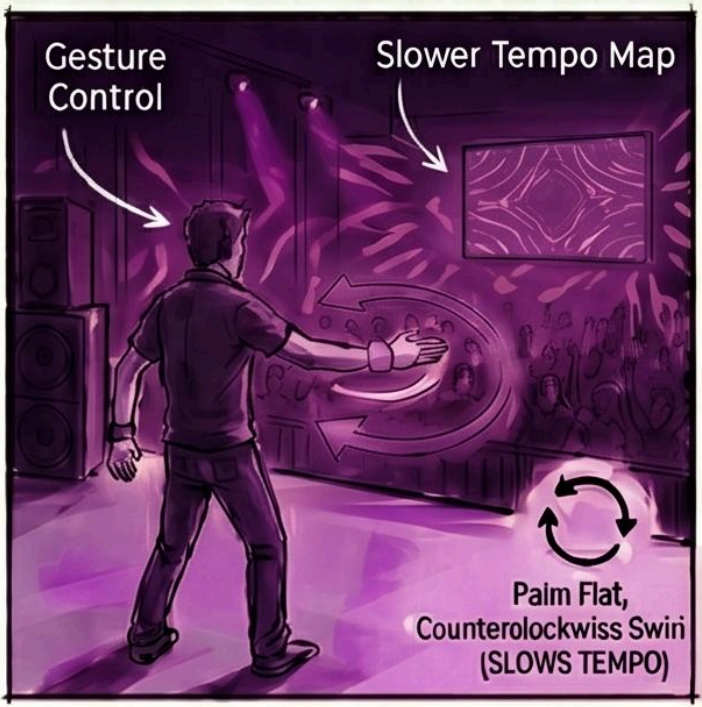
2. Haptic Conversion (Response)
- The intense bass and beat energy activate the sensor. The internal haptic engine translates acoustic energy into mechanical motion, precisely tracking the rhythm.



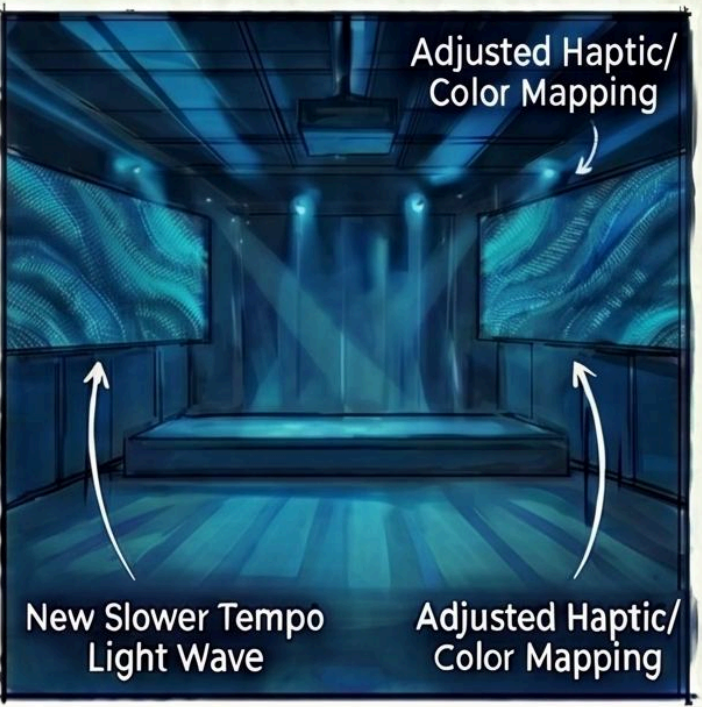
3. Haptic Pattern Mapping (Processing)
The real-time mapping system builds a detailed database of complex haptic patterns, categorizing and storing them based on rhythmic complexity and texture.



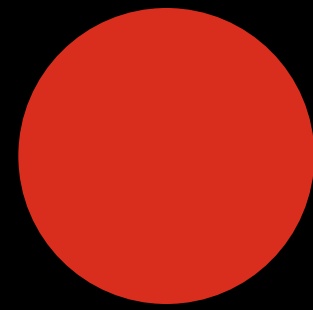
4. Gesture to Light Conversion (Output)
These stored haptic patterns are algorithmically translated into specific, complex cool light wave patterns and colors for the venue's illumination.



5. Tempo Change (Input)
The DJ performs a specific flat-palm, counterclockwise gesture over the mixer, which triggers the system to switch to a mapped, slower-tempo profile.



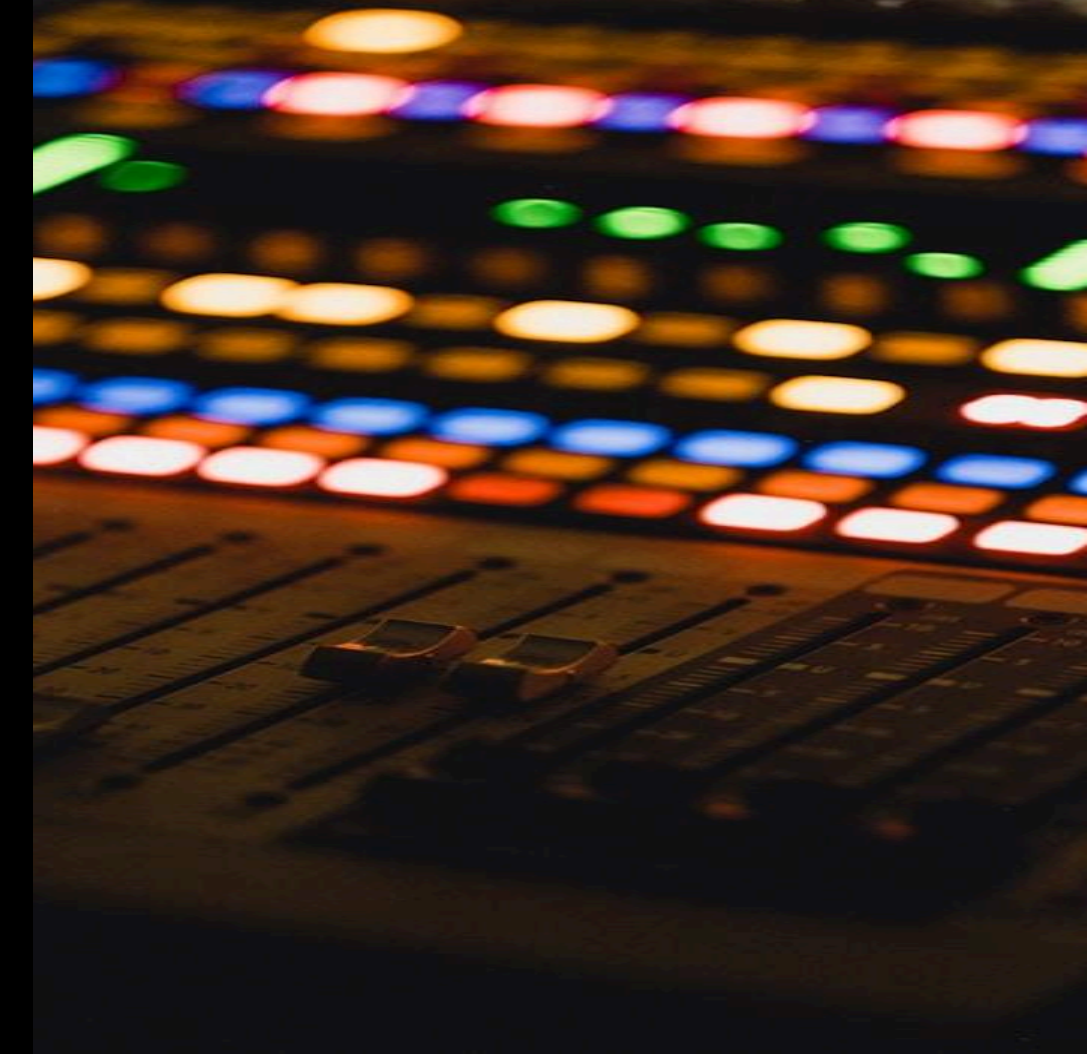
6. Adaptive Lighting (System Response)
The system instantly adapts the real-time lighting output. The light patterns, textures, and colors dramatically shift, reflecting the new slower tempo and the corresponding haptic profile, creating a whole new atmosphere.



Interaction specifications

SYSTEM 1:

Audio Gesture Control



● CORE IDEA

- Receive gesture input through a wearable device to translate into tempo changes

● GOAL

Give the DJ physical freedom on the stage to control the music tempo away from their work station.



Input System

Pressured Tap

Sensors used

Tactile Activation Button: Mounted on the side of the wearable wristband and detects users momentary push to capture and confirms user intention to activate gesture control.

Example: Push activation to enable audio gesture control.

Hand Gesture

Sensors used

Inertial Measurement Unit: Built inside a wearable wristband, the accelerometer, gyroscope, and magnetometer can detect wrist rotation, angular velocity, circular clockwise/counterclockwise motion, and gesture amplitude.

Example: Slow counter-clockwise hand movement for gradually slowing down the tempo.

Recognition Patterns



DEPRESSION FORCE (HOW HARD DOES THE BUTTON NEED TO BE PRESSED?)

PUSH → CURRENT MODE CONTINUES
RELEASE → CHANGE IN CONTROL MODE (BINARY GESTURE ON/OFF SWITCH), ACTIVATION RELIES ON THE BUTTON'S RELEASE ACTION

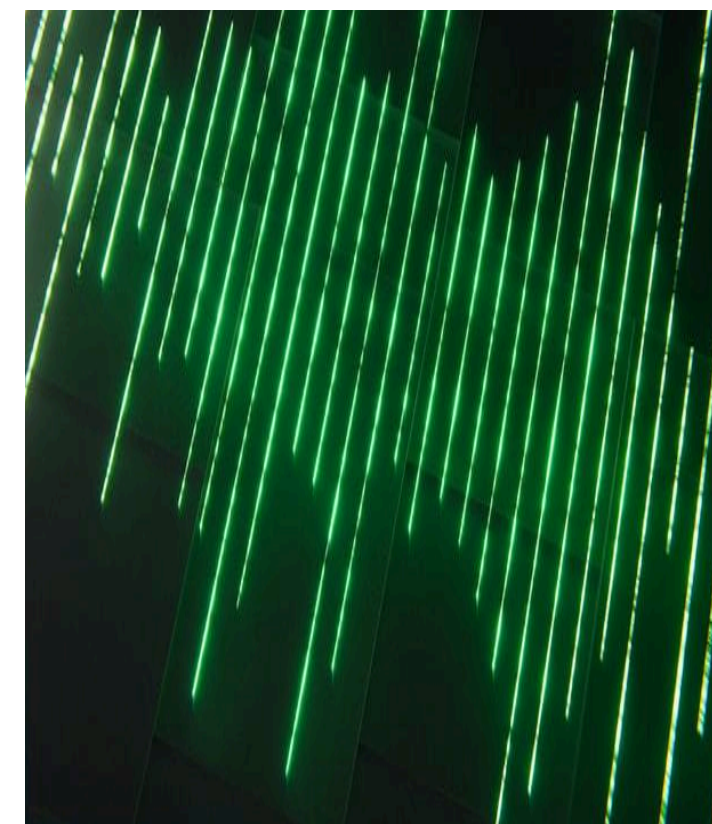
 ACTIVATE GESTURE CONTROL



MOVEMENT DIRECTION (HOW DOES DIRECTION AFFECT TEMPO CHANGES?)

CLOCKWISE → INCREASE IN BPM
COUNTER-CLOCKWISE → DECREASE IN BPM

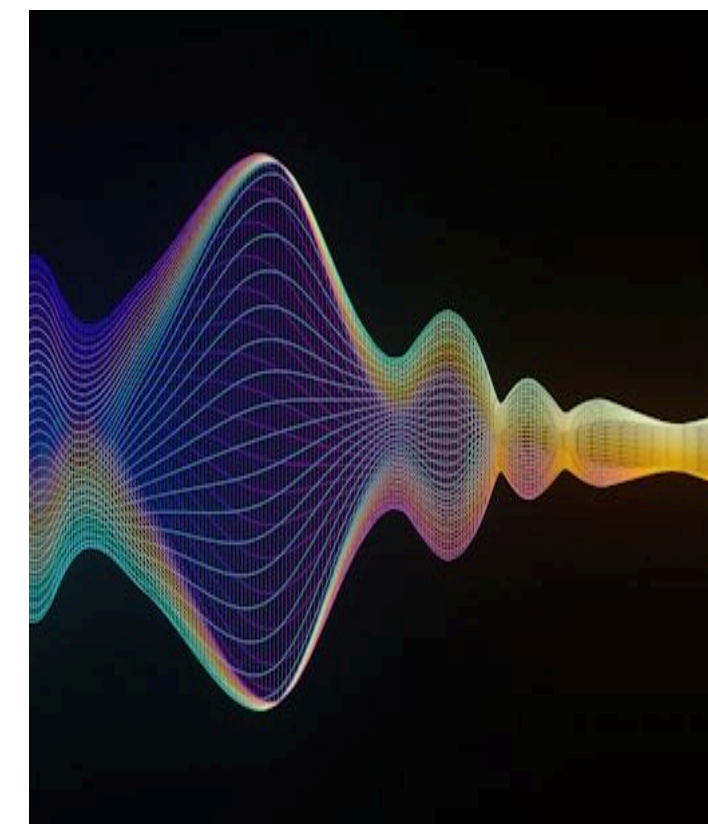
 +/- TEMPO CHANGE



SPEED (HOW DOES MOVEMENT SPEED AFFECT TEMPO CHANGES?)

NUMBER OF SECONDS → TIME FOR SPEED CHANGE

 SPEED OF TEMPO CHANGE



AMPLITUDE (HOW DOES SIZE OF MOVEMENT AFFECT THE TEMPO?)

QUARTER CIRCLE → +/- 0.25X ORIGINAL BPM
HALF CIRCLE → +/- 0.5X ORIGINAL BPM
FULL CIRCLE → +/- 1X ORIGINAL BPM

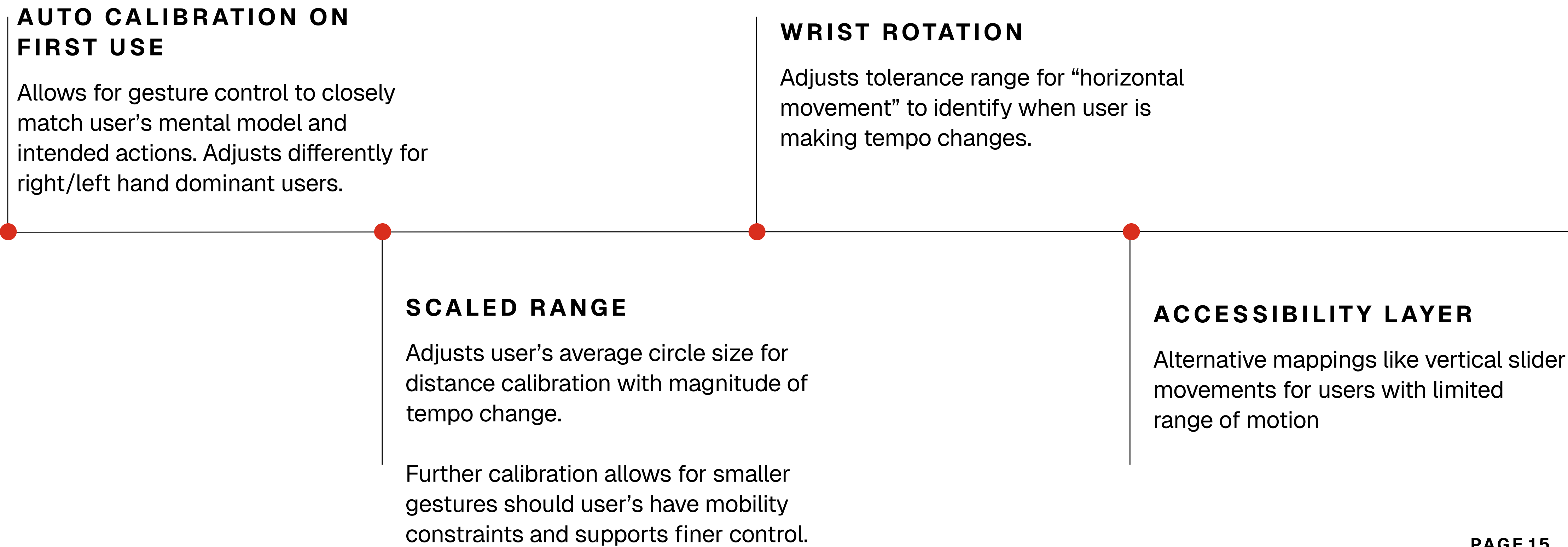
 % CHANGE IN BPM

Input → Output Mapping

INPUT FEATURE	DETECTED PARAMETER	OUTPUT MAPPING
Button Activation Press	Pressure	Gesture mode activation
Movement Angles	Horizontal Plane Changes	Gesture Validity
Motion Direction	Lateral Rotation Changes	BPM Increase/Decrease
Motion Diameter	Gesture Radius	Magnitude of BPM Shift
Motion Speed	Angular Velocity	Speed of Tempo Change

Adaption to User Variation

Not all users move identically. Adjusts for dominant hand and personal user movement range.



Outputs Produced

△ Tempo Change (BPM)

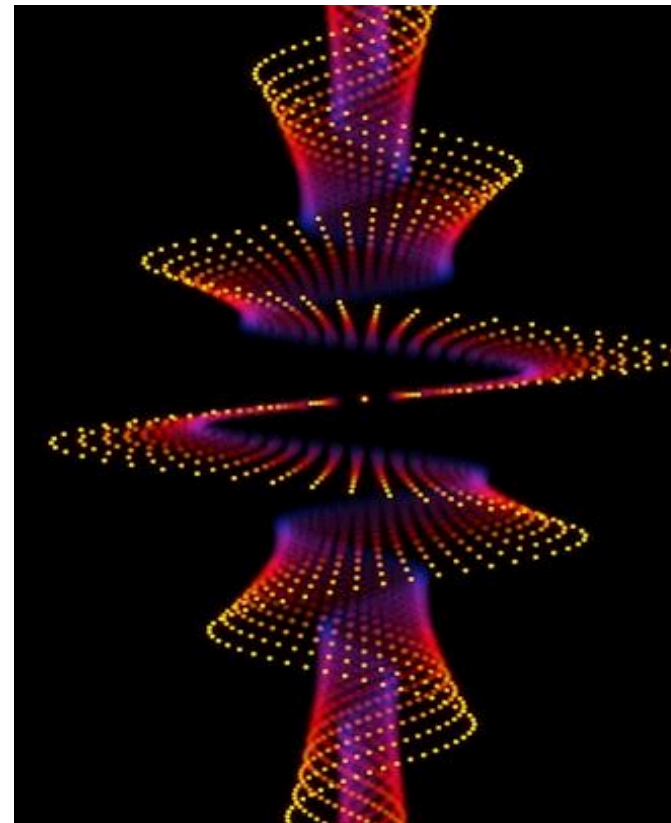
⌚ Speed of Tempo Change

Output Parameters



HAPTIC CLICK + MODE CHANGE

BUTTON PUSH/RELEASE → HAPTIC CLICK
HAPTIC CLICK → ON/OFF GESTURE MODE



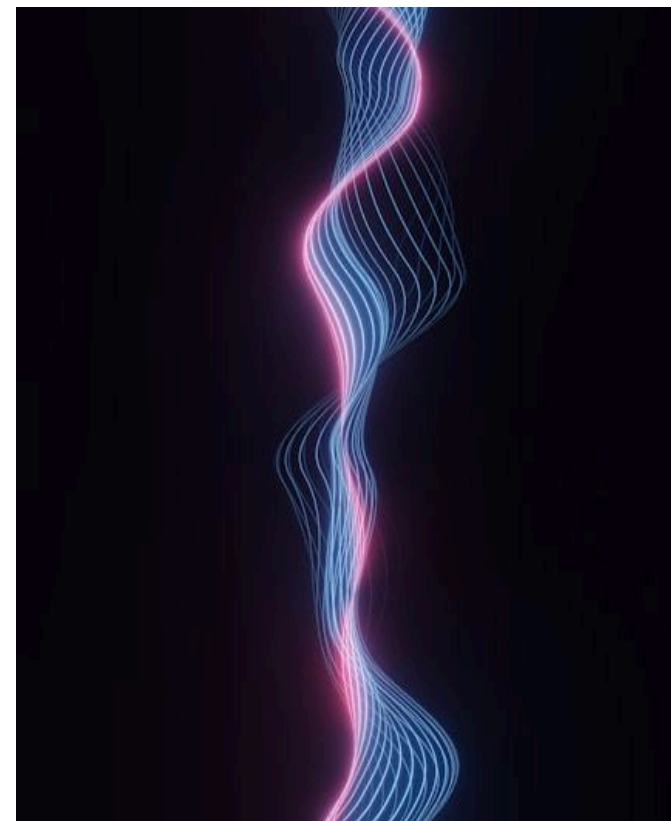
TEMPO INCREASE/DECREASE

ROTATIONAL DIRECTION:
• LEFT (COUNTER-CLOCKWISE) → DECREASE
• RIGHT (CLOCKWISE) → INCREASE



MAGNITUDE OF CHANGE

LINEAR MAPPING: 0-100% FOR ONE FULL
ROTATION



TRANSITION SPEED

• FAST MOVEMENT → RAPID TEMPO CHANGE
• SLOW MOVEMENT → GRADUAL TEMPO
CHANGE

Meaning Making

MEANING OF INPUTS (USER INTENT)

1. Button → User Activation
for physical freedom
2. Gesture Direction → Tempo
Increase/Decrease
3. Gesture Speed/Distance →
Magnitude/Speed of
Change

INTERACTION SUCCESS?

Auditory/Gesture Mapping: Consistent mapping (gesture control vs. traditional control) follow similar movements and auditory feedback helps reduce learning time to improve transitional ease between modes.

MEANING OF OUTPUTS (SYSTEM FEEDBACK)

1. Increase/Decrease Tempo
→ type of shifts in music
2. Speed of Change → quick/
slow energy shifts



Handling Ambiguity

Movement vs Intention

1. Uses a gyroscopic sensor to determine lateral movement before further assessing directional inputs
2. If movement falls within the tolerance range for horizontal plane asses based on other factors whether to reject or accept gestures
3. Prevents accidental changes in BPM

Gesture Confidence Levels

1. Follows user's calibrated circular range → enable controls
2. Lack of directional consistency (movement along the circle circumference) prevents unintentional gesture activation
3. Without a >85% confidence based on movement factors, gestures are rejected

Button Activation

1. Mode activation located on wearable to enable user to activate gesture control at any point
2. Button switch controls when movements are being read to prevent overlap with traditional music mixing movements.

SYSTEM 1:



INPUT

Tap +
gesture
control



OUTPUT

Dynamic audio
(tempo changes)

SYSTEM 2:

Resonant Club Lighting System

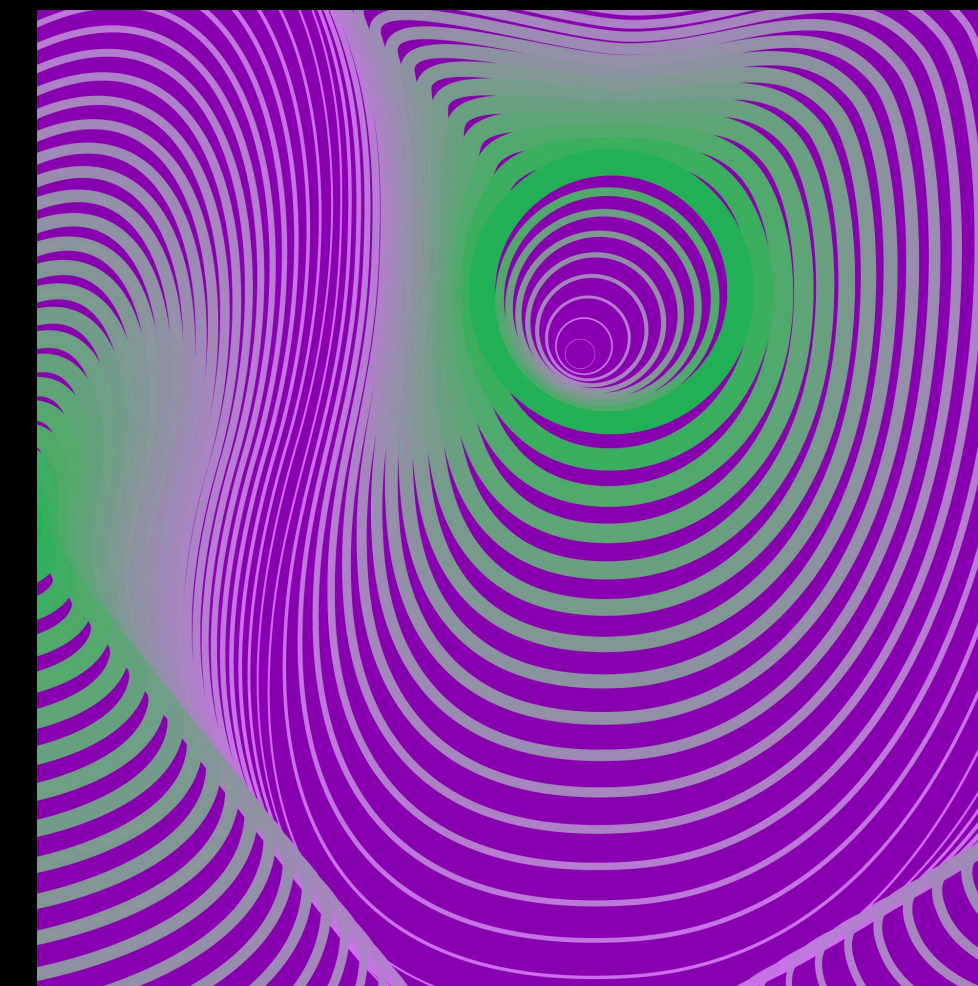


● CORE IDEA

- Convert audio + vibration (previously output) into primary inputs
- Translate into dynamic lighting environment

● GOAL

Create a multi-sensory, immersive club environment where sound becomes visible



Input System

🔊 Audio Signals

Sensors used

Directional Microphone Array: Captures airborne audio and enables frequency + amplitude analysis. Detects sound waves traveling through air. Detects sound waves traveling through air.

Example: Vocals (mid–high frequencies), Instrumentation, Tonal characteristics (pitch, timbre, sharpness)

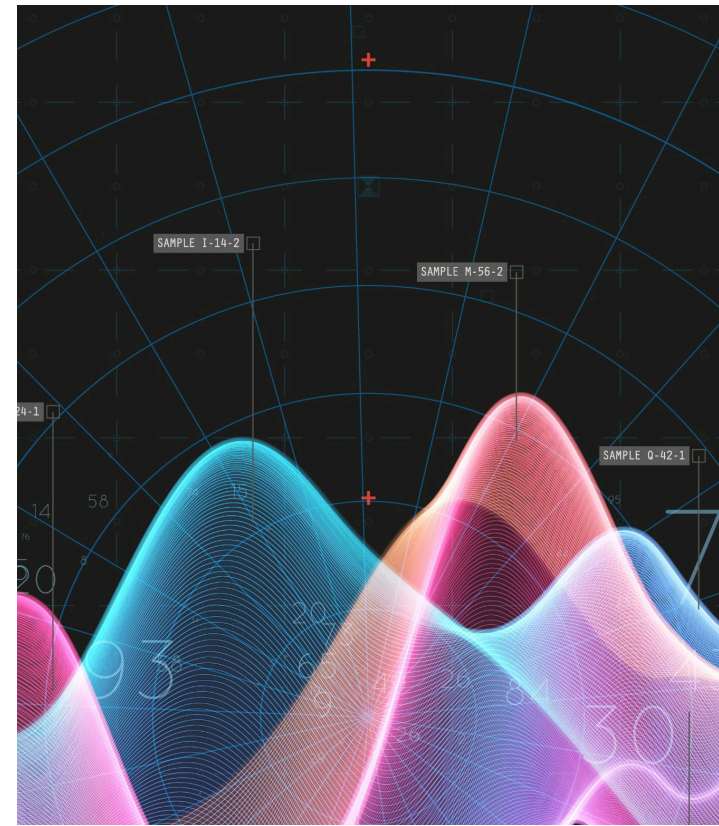
🔊 Vibrations

Sensors used

Contact Piezoelectric Vibration Sensor: Mounted on speaker surface and can detect mechanical vibrations (20 Hz – 200 Hz bass range). Detects mechanical vibrations directly from the speaker surface.

Example: Beat impact (kick drum, bass drops)

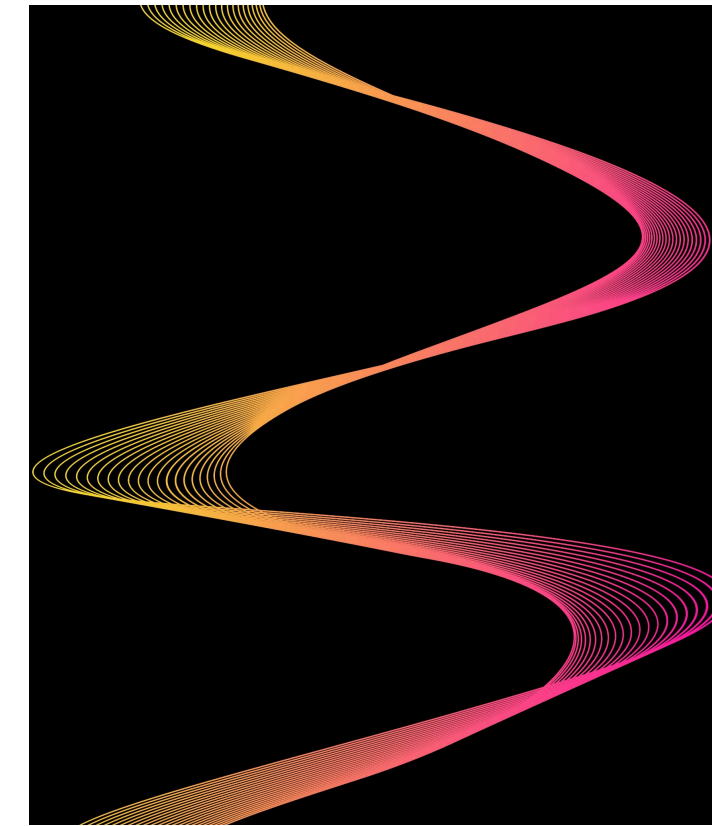
Recognition Patterns



FREQUENCY (WHAT KIND OF SOUND IS IT?)

LOW (20–250 HZ) → BASS (KICK, DROPS)
MID (250–2000 HZ) → VOCALS, INSTRUMENTS
HIGH (2000–20,000 HZ) → SHARP SOUNDS

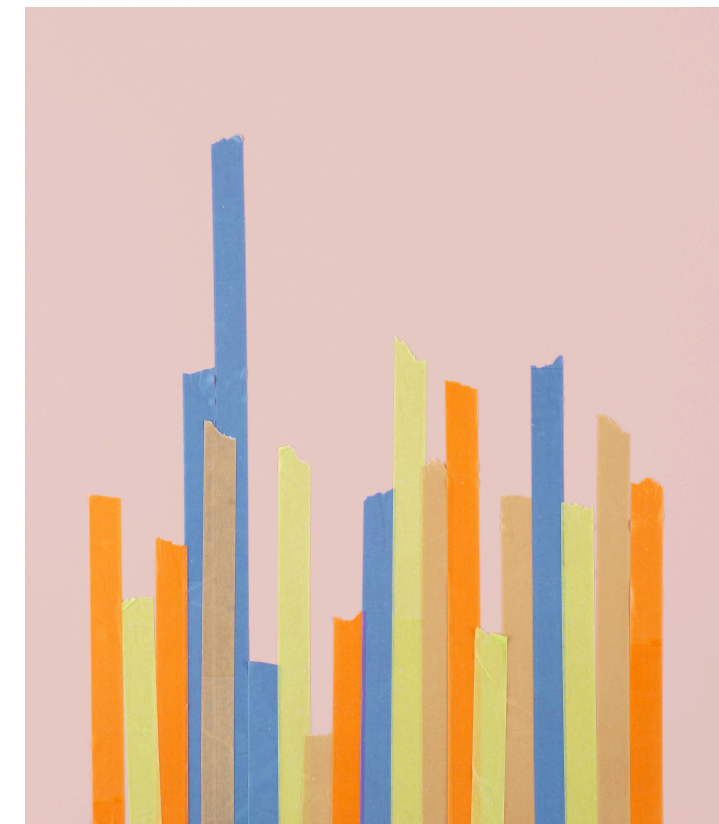
💡 COLOR HUE CHANGE OF LIGHT



BEAT / RHYTHM (IS THERE A REPEATING PATTERN?)

60–180 BPM (BEATS PER MINUTE)

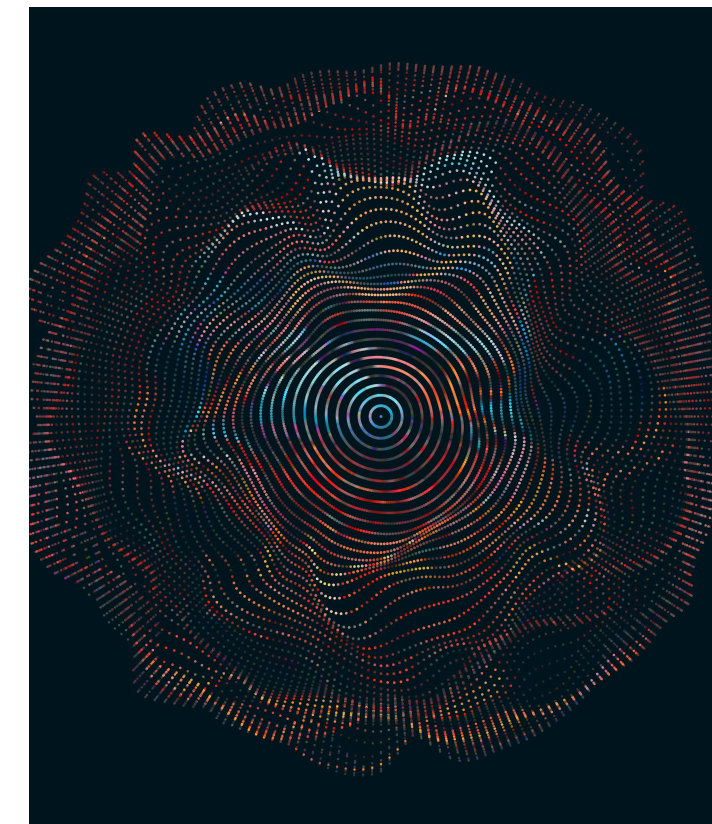
💡 BLINKING / PULSING OF LIGHTS



LOUDNESS (AMPLITUDE) (HOW STRONG IS THE SOUND?)

LOW (<60 DB) → CALM / BACKGROUND
MEDIUM (60–85 DB) → NORMAL MUSIC LEVEL
HIGH (>85 DB) → PEAK / DROP

💡 BRIGHTNESS OF LIGHT



VIBRATION PATTERN (HOW DOES THE SPEAKER MOVE?)

CONTINUOUS VIBRATION → STEADY SOUND
PULSED VIBRATION → BEATS
IRREGULAR VIBRATION → NOISE / CROWD

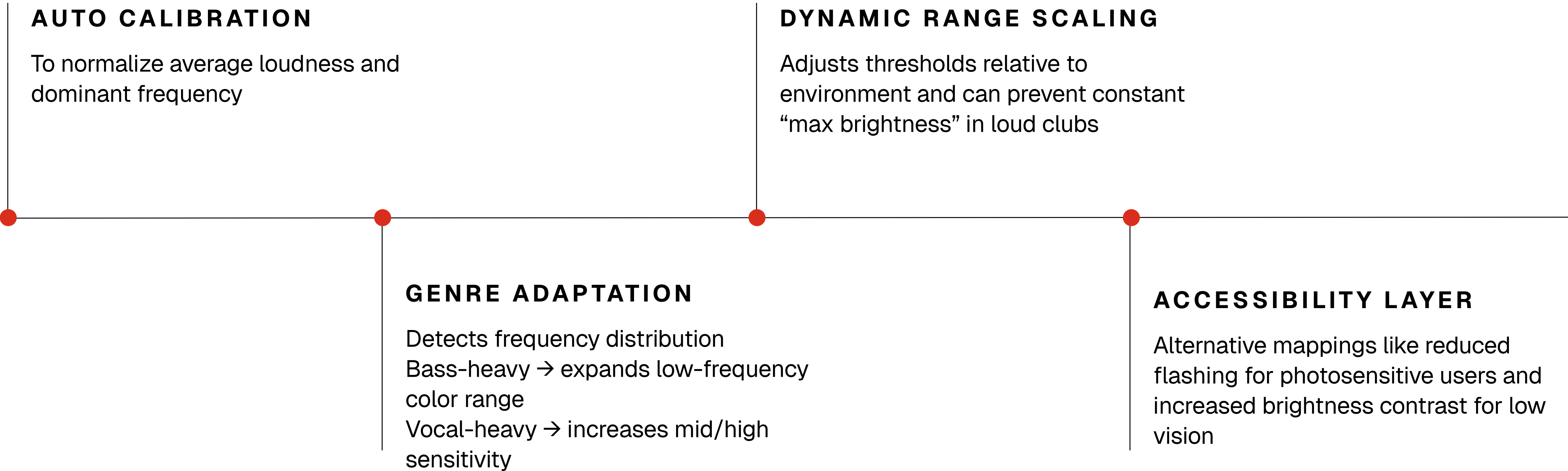
💡 RHYTHM DETECTION & INTENSITY CHANGES OF LIGHT

Input → Output Mapping

INPUT FEATURE	DETECTED PARAMETER	OUTPUT MAPPING
Frequency	Hz band	Color (hue)
Amplitude	dB level	Brightness
Beat	BPM	Blink rhythm
Spectral sharpness	centroid	Light sharpness / transition speed
Vibration pattern	waveform type	Pattern complexity

Adaptation to User Variation

Club environments are highly variable: Different music genres, Speaker setups and crowd density



Outputs Produced

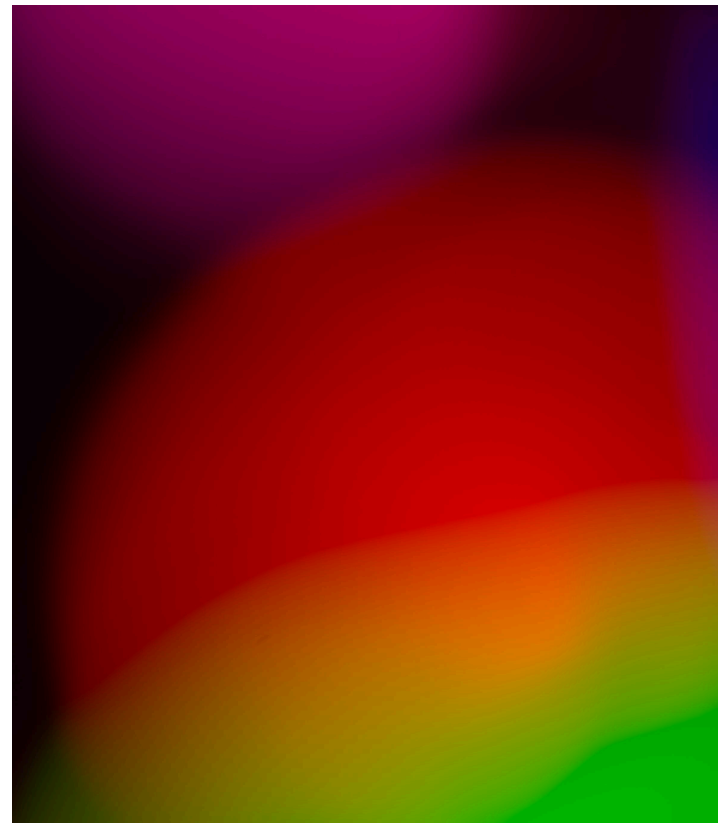
🔴 Color change (hue)

☀️ Brightness variation

🕒 Blinking light pattern

⚡ Transition sharpness

Output Parameters



FREQUENCY → HUE MAPPING:

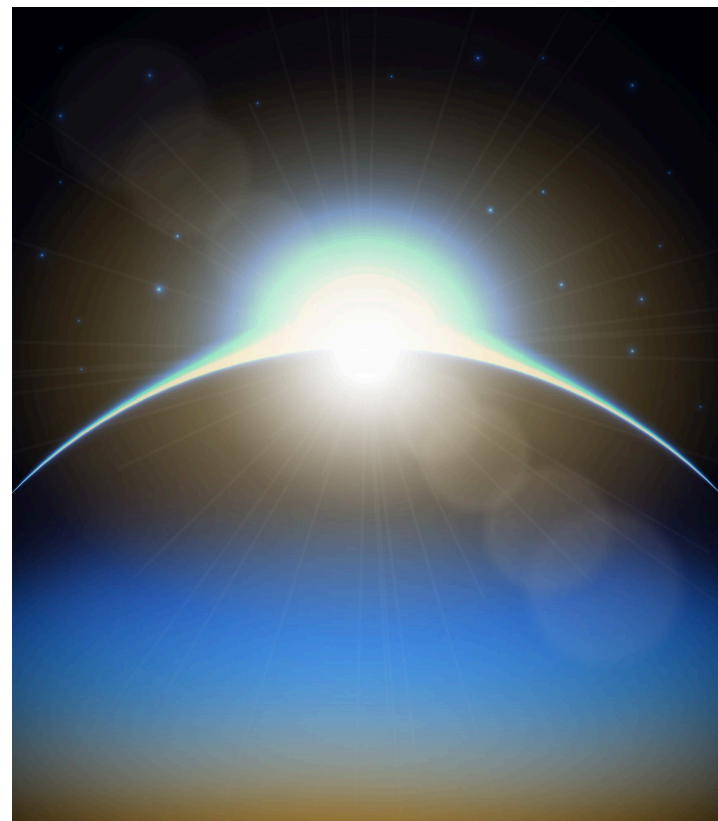
20–250 HZ → RED / ORANGE
250–2K HZ → GREEN / YELLOW
2K–20K HZ → BLUE / VIOLET



BLINK PATTERN (RHYTHM)

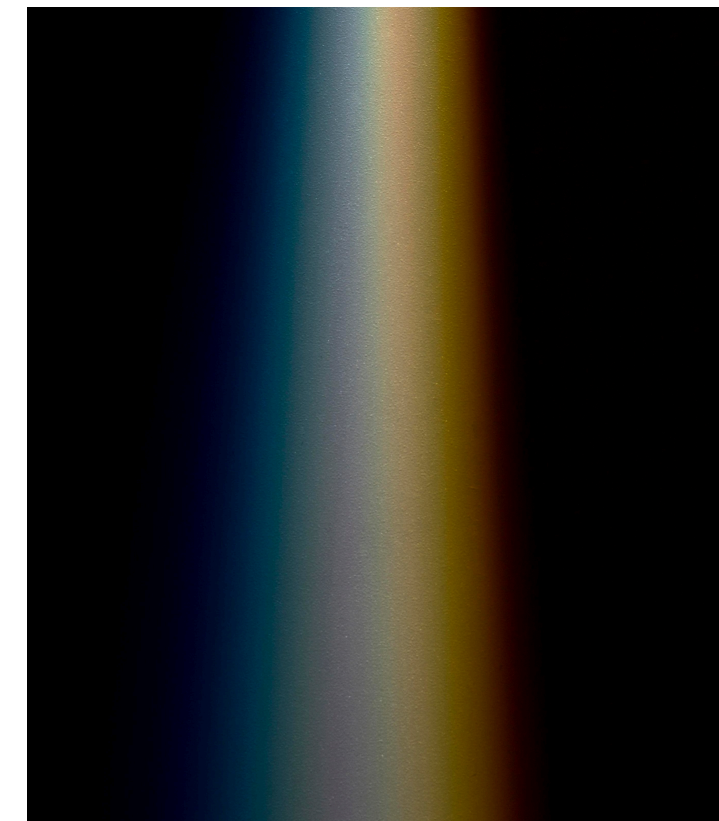
BPM → PULSE RATE:

- 60 BPM → 1 HZ PULSE
- 120 BPM → 2 HZ PULSE



BRIGHTNESS (INTENSITY)

LINEAR MAPPING: 0–100% BRIGHTNESS
BASED ON NORMALIZED DB



SHARPNESS / TRANSITION SPEED

- HIGH FREQUENCY → SHARP, FAST TRANSITIONS (~50MS)
- LOW FREQUENCY → SMOOTH FADES (~300–800MS)

Meaning Making



MEANING OF INPUTS (USER INTENT)

1. Music → Controls atmosphere
2. Bass drops → Signal climax moments
3. Rhythmic patterns → Crowd synchronization

INTERACTION SUCCESS?

1. Immediate synchronization: Lights match beat within <50ms latency
2. Visual coherence: Consistent mapping (same sound → same color)



MEANING OF OUTPUTS (SYSTEM FEEDBACK)

1. Brightness → energy level
2. Color → mood
3. Rhythm → beat alignment
4. Sharpness → intensity



Handling Ambiguity

Overlapping Frequencies

1. Uses a priority order: beat first, then loudness, then frequency
2. If nothing stands out clearly, it avoids reacting
3. Falls back to a neutral light instead of mixing signals

Crowd Noise Interference

1. Ignores random sounds like shouts or sudden spikes
2. Smooths the signal to filter out background noise
3. Only reacts when the input is strong and consistent

Irregular Beats

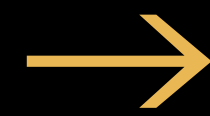
1. Smooths out the rhythm to avoid jumpy lighting
2. If the beat isn't clear, it stops trying to sync
3. Switches to a slow pulse instead of erratic flashing

SYSTEM 2:



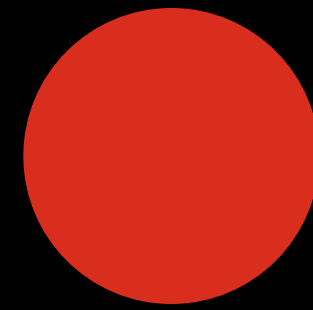
INPUT

Sound +
haptic
vibrations



OUTPUT

Dynamic visual
lighting (color,
brightness, pulse)



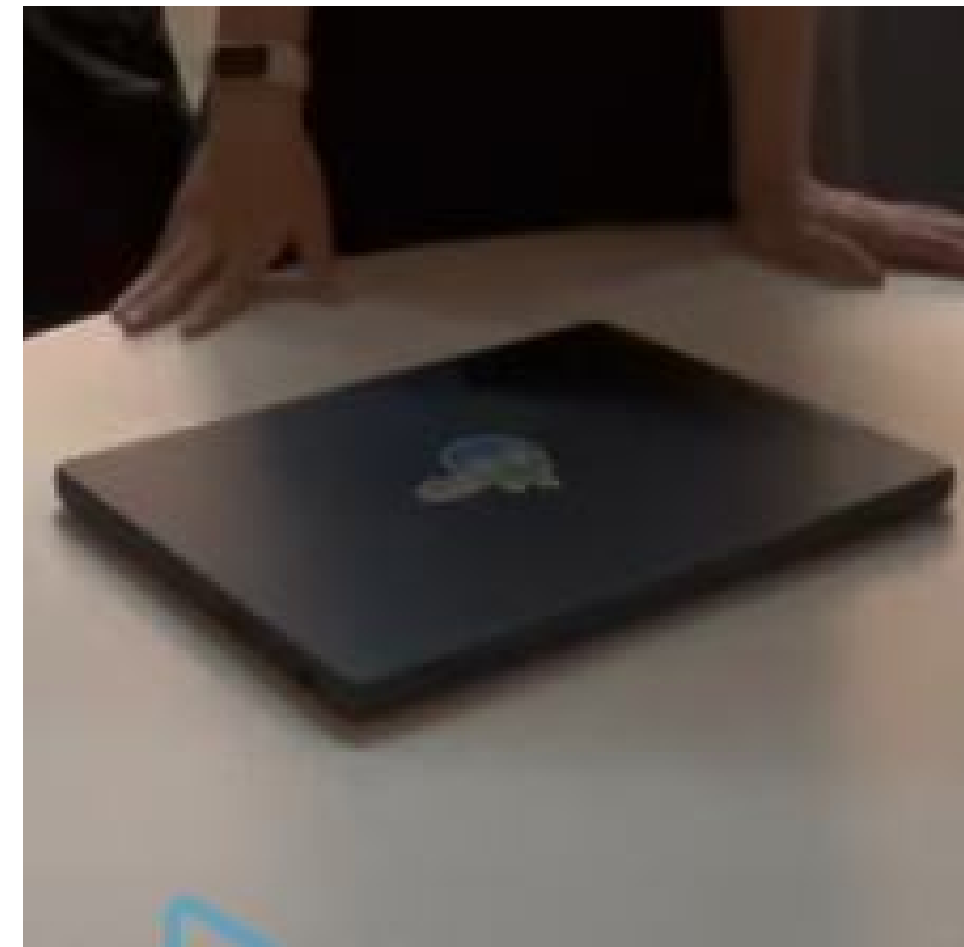
Building the Wizard-of-Oz Prototype

Prototype Components: Hardware



● WRIST BAND

A Fitbit was used to represent the wrist band that detects hand gesture.



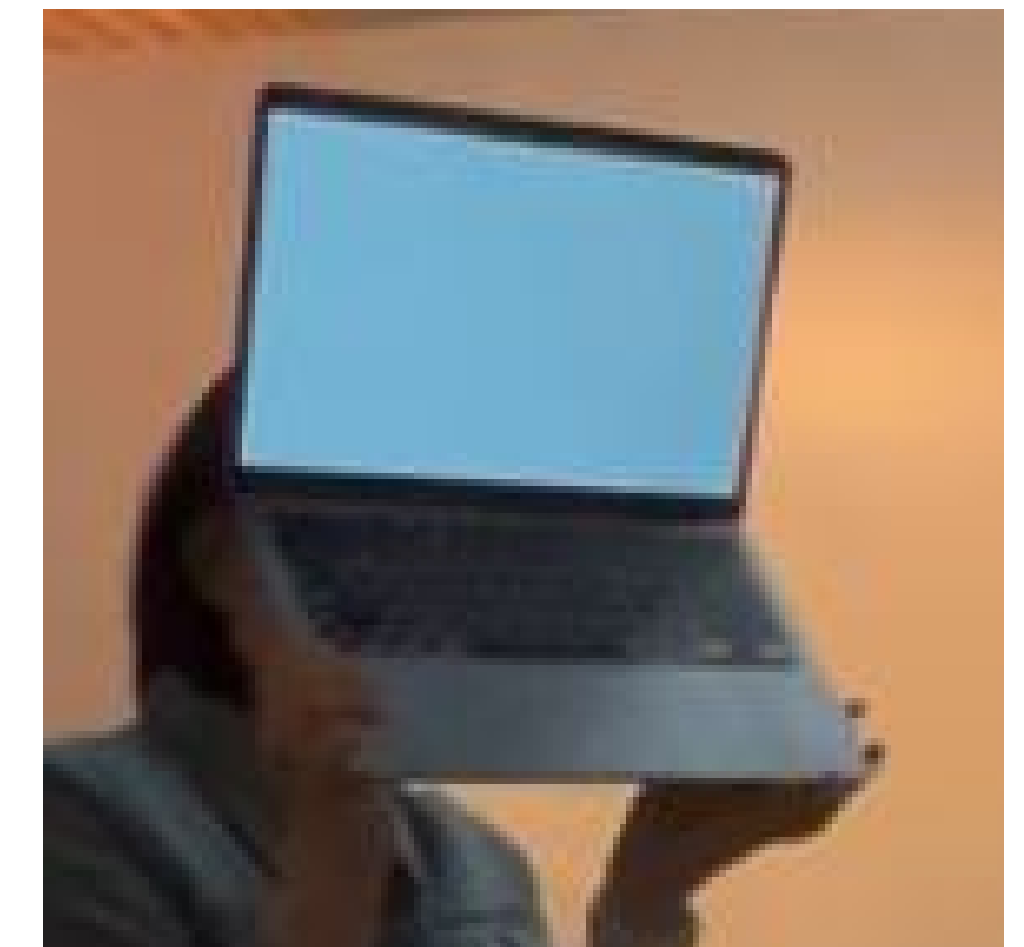
● DJ'S MIXER

A MacBook was used as a stand-in for the DJ's mixer where DJ can turn on the AirDJ. We used the sticker in the middle of the device as the activation button.



● SPEAKERS

A water bottle and a tumbler was used to represent speakers in the simulation.



● LIGHT SYSTEM

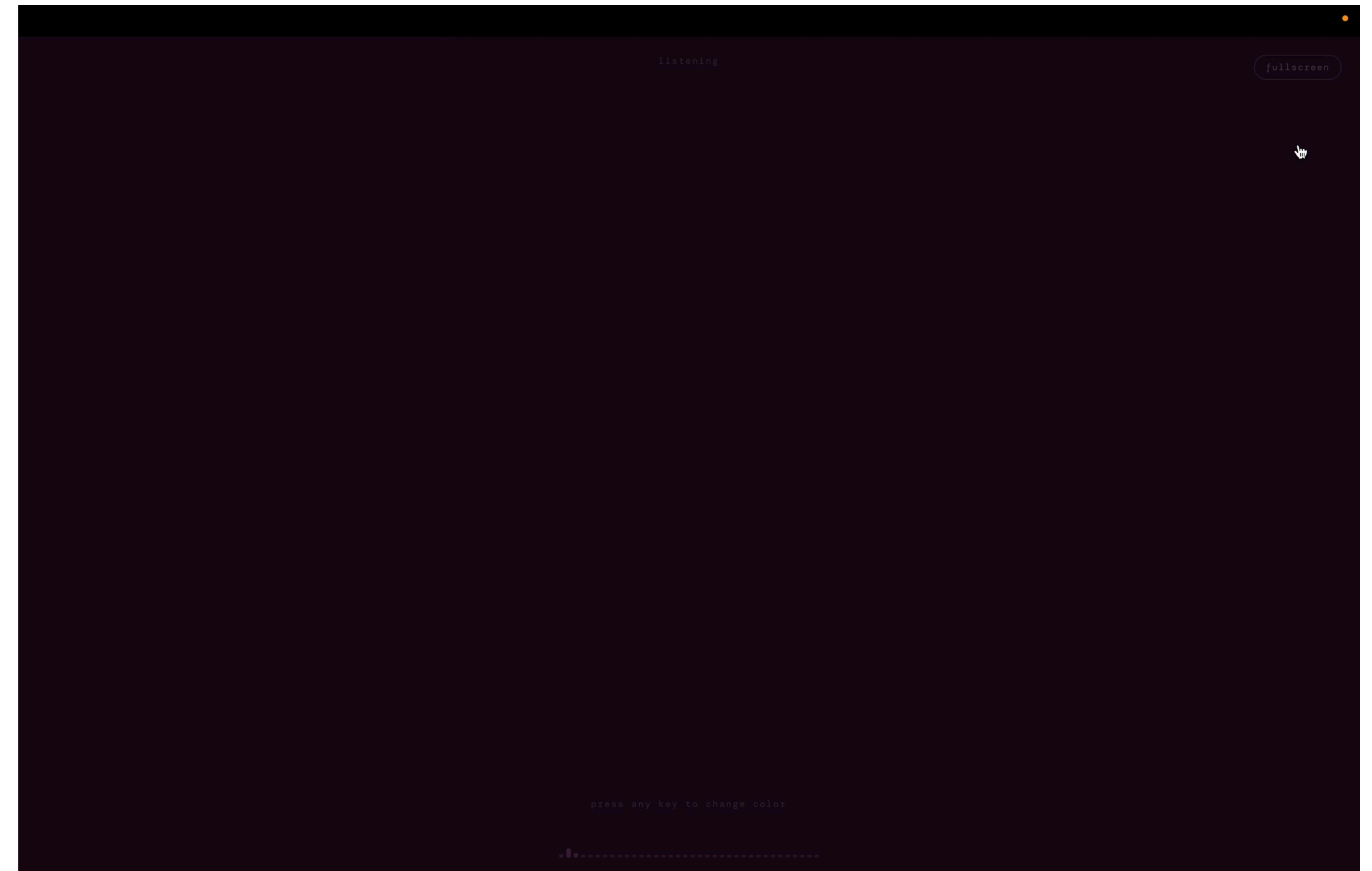
A MacBook screen was used to display the light system behind the user; in V2, we plan to use an actual projector.

Prototype Components: Software

To simulate the sound-reactive lighting system described in the previous slides, a prototype was vibe-coded via Claude.

References:

- Prototype file: [sound-light-v3.html](#)
- Music used in the screen recording: [Yaeji | Boiler Room: New York \(24:40-25:42\)](#)



Prototyping Process

01

Lina, joining the team remotely via Zoom, vibe-coded the prototype of the sound-reactive lighting system using Claude.

02

A variety of props were gathered and placed on a table to represent the AirDJ screenless DJ performance system.

03

The team split roles to 'Wizard-of-Oz' the AirDJ experience:

- YeaRim held the MacBook behind the DJ to demonstrate how the lighting system would react to the music and be displayed to the crowd.
- Janny controlled the tempo of the music using the playback speed controller in the Youtube video player, following Nara's hand gestures.
- Rachit recorded the testing session.

04

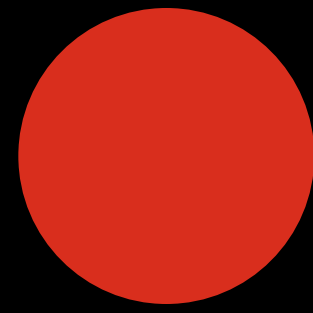
Nara, playing the role of the DJ, experienced the simulation.

Wizard-of-Oz Testing



Using the props and the light system software prototype, we conducted our Wizard of Oz testing session to observe how our designed interactions would unfold in real life.





Testing Results

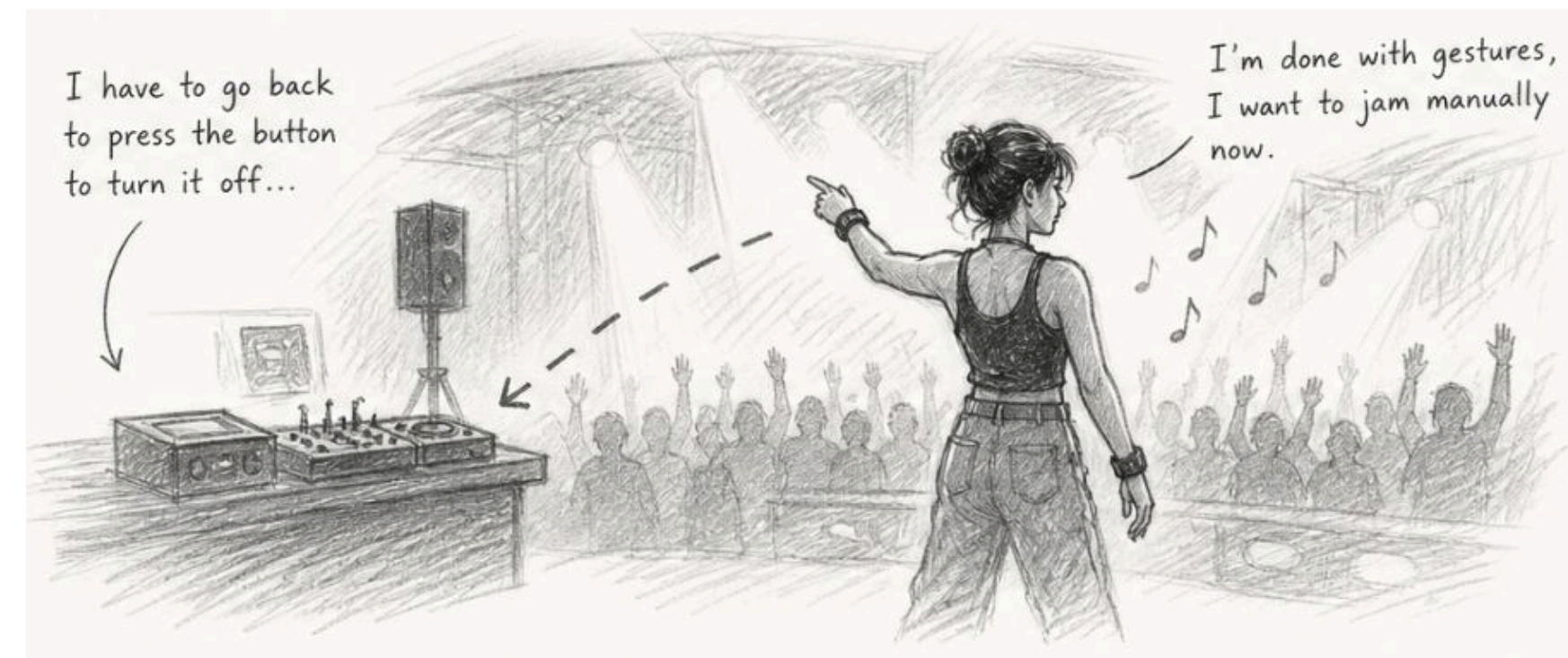
What the User Did, and What Confused Her

01



When the DJ started slowing down the music, she was confused about the range of the hand motion and how it related to the tempo. The tempo change did not correspond to her gestures.

02



After using hand gestures to control the music, the DJ wanted to switch back to manual control and jam freely.

However, she realized she must physically return to the station to press the button and turn off the AirJD mode. She felt this interrupted the flow and pulled her away from the audience.

Which Outputs Communicated Clearly / Which Were Missed or Misread

The lighting changed according to the beat and the volume of the music, which was easy to recognize.



What should be changed in V2?

We identified several technical and interaction issues from the first test run that we want to improve in the next iteration:

- **More noticeable lighting feedback**

The lighting response needs to be more visually obvious so the DJ can immediately perceive how their hand gestures are affecting the music tempo and the surrounding atmosphere.

- **Smoother tempo transitions during gesture control**

During the first prototype test, tempo changes felt too abrupt when the DJ attempted to slow down the music gradually. This happened because the YouTube playback speed control does not allow continuous adjustments.

For the next Wizard of Oz prototype, the operator behind the scenes will manually adjust the tempo in smaller incremental steps, continuously matching the DJ's hand movement until the gesture stops. This should create a smoother and more natural interaction.



What should be changed in the Product Itself?

More standardized gesture definitions:

- Gesture controls need to be more clearly defined and standardized to reduce ambiguity and confusion during performance. Each gesture should have a consistent motion range, direction, and corresponding system response so DJs can better predict the outcome of their actions.

Move activation button from the DJ mixer to the wearable

- After the first test run, we found that it is difficult for the DJ to move back to the mixer and turn off the button that activates air gesture. We decided to put the button to the wearable instead.



AI Usage

- 
- Claude and Gemini were used to check for grammar and improve writing clarity.
 - Gemini was used to create storyboards.
 - Claude was used to vibe-code the lighting system prototype.

● THANK YOU!